Equations, Ratio & Similarity

PRACTICE SET

Prof. Fang Math

Name: _____ Date: _____

Each topic starts with a **worked example**. Study it, then solve the problems below it. Show your work in the space given.

1. Solving Equations

KEY IDEA

Undo the operations in reverse order. Whatever you do to one side, do to the other side.

WORKED EXAMPLE

Solve $3x - 7 = 11$.

Step	Why
$3x - 7 + 7 = 11 + 7$	Add 7 to both sides (undo the subtraction).
$3x = 18$	Simplify.
$\frac{3x}{3} = \frac{18}{3}$	Divide both sides by 3 (undo the multiplication).
$x = 6$	Check: $3(6) - 7 = 11$ ✓

Your turn

1. EASY $x + 9 = 23$

2. EASY $6x = 42$

3. EASY $\frac{x}{4} + 3 = 8$

4. MEDIUM $5 - 2x = 17$

5. HARD $\frac{2}{3}x - 5 = 7$

© 2026 Prof. Fang Math. All rights reserved.

2. Multi-Step Equations

KEY IDEA

Simplify each side first (distribute, combine like terms). Then get the variable on one side and solve.

WORKED EXAMPLE

Solve $4(x-2)+3=2x+9$.

Step	Why
$4x-8+3=2x+9$	Distribute the 4.
$4x-5=2x+9$	Combine like terms.
$2x-5=9$	Subtract $2x$ from both sides.
$2x=14 \quad \Rightarrow \quad x=7$	Add 5, then divide by 2. Check: $4(5)+3=23$ and $2(7)+9=23$ ✓

Your turn

6. EASY $2(x+3)=14$

7. EASY $3x+2x-4=21$

8. EASY $4x+1=2x+9$

9. MEDIUM $3(2x-1)=4(x+2)+1$

10. HARD $\dfrac{3x-1}{4}=\dfrac{x+5}{2}$

© 2026 Prof. Fang Math. All rights reserved.

3. Literal Equations

KEY IDEA

A literal equation has several letters. To solve for one letter, treat the other letters like numbers and use the same steps.

WORKED EXAMPLE

Solve $P = 2l + 2w$ for l.

Step	Why
$P - 2w = 2l$	Subtract $2w$ from both sides.
$\frac{P - 2w}{2} = l$	Divide both sides by 2.
$l = \frac{P - 2w}{2}$	Check with numbers: $l = 5, w = 3 \Rightarrow P = 16$, and $\frac{16 - 6}{2} = 5$ ✓

Your turn

11. EASY Solve $d = rt$ for t.

12. EASY Solve $A = lw$ for w.

13. EASY Solve $y = x + b$ for b.

14. MEDIUM Solve $y = mx + b$ for x.

15. HARD Solve $A = \frac{1}{2}h(b_1 + b_2)$ for b_2.

© 2026 Prof. Fang Math. All rights reserved.

4. Ratio and Proportion

KEY IDEA

A ratio compares two quantities ($a : b$ or $\frac{a}{b}$). A proportion says two ratios are equal; solve it by cross-multiplying.

WORKED EXAMPLE

A recipe uses 3 cups of flour for every 2 cups of milk. How much flour goes with 8 cups of milk?

Step	Why
$\frac{3}{2} = \frac{x}{8}$	Set up a proportion: flour over milk on both sides.
$2x = 3 \cdot 8$	Cross-multiply.
$2x = 24 \quad \Rightarrow \quad x = 12$	Divide by 2.
12 cups of flour	Check: $12 : 8 = 3 : 2$ ✓

Your turn

16. EASY Write the ratio $15 : 25$ in simplest form.

17. EASY Solve the proportion $\frac{x}{5} = \frac{12}{20}$.

18. EASY 6 notebooks cost $18. What is the cost of 1 notebook (the unit rate)?

19. MEDIUM The ratio of boys to girls in a club is $3 : 5$. There are 64 students in all. How many are girls?

20. HARD A bag has red and blue marbles in the ratio $3 : 4$. After 6 more red marbles are added, the ratio becomes $1 : 1$. How many blue marbles are in the bag?

© 2026 Prof. Fang Math. All rights reserved.

5. Similar Figures

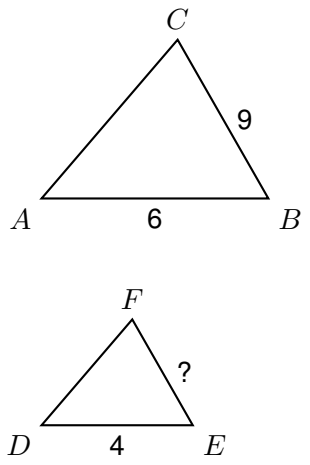
KEY IDEA

Similar figures ($\sim$) have the same shape: matching angles are equal and matching sides are in the same ratio (the scale factor).

WORKED EXAMPLE

$\triangle ABC \sim \triangle DEF$ with $AB = 6$, $BC = 9$ and $DE = 4$. Find EF.

Step	Why
$\frac{DE}{AB} = \frac{EF}{BC}$	Matching sides are proportional (use the order of the letters).
$\frac{4}{6} = \frac{EF}{9}$	Substitute.
$6 \cdot EF = 36$	Cross-multiply.
$EF = 6$	Divide by 6. Check: $\frac{4}{6} = \frac{6}{9} = \frac{2}{3}$ ✓



Your turn

21. **EASY** $\triangle ABC \sim \triangle DEF$, $AB = 5$ and $DE = 15$. What is the scale factor from $\triangle ABC$ to $\triangle DEF$?

22. **EASY** $\triangle ABC \sim \triangle DEF$ and $m\angle A = 40°$. What is $m\angle D$?

23. **EASY** A 3×5 rectangle is similar to a rectangle whose short side is 6. How long is its long side?

24. **MEDIUM** A 6-foot-tall person casts a 4-foot shadow. At the same time, a tree casts an 18-foot shadow. How tall is the tree?

25. **HARD** Two similar triangles have a scale factor of $2 : 3$ (small to large). The small triangle has area 20 square units. Find the area of the large triangle.

© 2026 Prof. Fang Math. All rights reserved.